

Matteo Cherubini

Front-end developer • Applied AI and knowledge systems

Nationality Italian

Place of birth Florence, Italy

Phone (+39) 327 6615664

GitHub github.com/MatteoCherubini

Date of birth 27 March 1997

Address via Santa Gonda 108, 59100 Prato (PO), Italy

Email cherubinimatte@gmail.com

Medium medium.com/@cherubinimatte

ABOUT ME

Front-end developer (Angular) with three years' experience building the digital backbone of Italian education and cultural heritage: HUB Scuola, the main learning platform for Italian schools, and platforms for the National Central Library of Rome and CNR research infrastructures linked to the European Open Science Cloud — with a consistent focus on web accessibility.

Now an independent consultant in applied AI and knowledge management, designing systems end-to-end, from the security model to local LLM inference and human review. I work equally well inside structured teams and in full autonomy, including projects for regional public bodies run from requirements to delivery. C1 English, certified.

My method is algorithmic: break the problem down, optimise the flow, keep the map in my head, and write it down so it outlives me.

WORK EXPERIENCE

24 Jan 2023 – 1 May 2026

Florence and Pisa, Italy

Front-end developer

Gruppo Meta / M.E.T.A. S.r.l. (ETT Group), later Deda Next S.r.l. (Dedagroup)

Front-end developer (Angular) in a software house specialised in the digitalisation of educational publishing, cultural heritage and research. Permanent contract; work split between the Florence and Pisa offices.

- Angular development for Mondadori Education's HUB Scuola, the main digital learning platform for Italian schools: rollout of the new design system and independent development of new features, in particular interactive quizzes and exercises embedded in e-books.
- Translation of design mock-ups into Angular components; design and implementation of entire application flows of varying complexity.
- Web accessibility (a11y) in line with EU Directive 2016/2102; refactoring, debugging and unit testing.
- Large-scale modernisation of legacy viewers for complex editorial formats, carried out autonomously in close cooperation with the design and UX teams.
- Front-end contribution, within project teams, to platforms for national cultural and research institutions: digitalisation of the collections of the National Central Library of Rome, and CNR research infrastructures in the digital humanities (funded under NextGenerationEU and linked to the European Open Science Cloud).
- Projects for regional public bodies managed end-to-end, from requirements gathering to delivery, with direct contact with the client.
- Development and integration of Node.js APIs together with the back-end team; data visualisation (charts and tables).
- Tools: Angular, TypeScript, Node.js, Docker, Postman, Figma, Git (structured repositories with submodules).

1 May 2026 – Current
Prato, Italy

ICT consultant

Self-employed

Independent technology consultant working on applied AI and knowledge management.

- Integration of AI-based tools for professional firms (accountancy practices), automating repetitive workflows to free human time for analysis and judgment.
- Design and development of AI systems: OCR pipelines, self-maintaining documentation, retrieval architectures beyond the standard RAG approach.
- Design and operation of a self-hosted infrastructure supporting this work: containerisation, virtualisation, networking, security and local LLM inference on GPU.
- See Projects below for the systems built in this period.

EDUCATION AND TRAINING

1 Oct 2016 – 7 Jun 2022

Florence, Italy [EQF 6](#)

Laurea in Informatica (Bachelor's Degree in Computer Science) – Class L-31, Computer Science and Technologies

University of Florence – School of Mathematical, Physical and Natural Sciences

Three-year degree in Computer Science (180 ECTS).

- Programming and software engineering: programming, programming methodologies, concurrent programming, algorithms and data structures, interpreters and compilers, theoretical computer science.
- Systems and networks: operating systems, computer architecture, computer networks.
- Data and mathematics: databases and information systems, probability and statistics, numerical analysis, calculus I and II, linear algebra, discrete mathematics and logic, general physics.
- Security: computer forensics – digital investigation and the legal framework of IT evidence.
- Design and business: multimedia design and production, business skills.
- Curricular internship (12 ECTS).
- Final thesis: “Progressive Web Applications and Web Operating Systems” (supervisor: Prof. Rosario Pugliese; co-supervisor: Prof. Francesco Tiezzi). An analysis of Progressive Web Apps as a standards-based, cross-platform alternative to native applications, including a fully working PWA developed with Node.js, Ionic, Vue.js and Firebase.

22 Aug 2022 – 12 Oct 2022

Indianapolis, United States

Program for Intensive English (PIE) – Level 6 of 7, corresponding to CEFR C1

Indiana University–Purdue University Indianapolis (IUPUI)

Intensive academic English programme (autumn session) at an American university.

- Achieved level 6 out of 7 in both Reading–Writing and Listening–Speaking, corresponding to CEFR level C1.
- Academic writing, oral presentation and discussion skills.
- Full immersion in an international student environment.

28 Oct 2025 – Current

Prato, Italy

Sommelier training course – Level 1 completed, Level 2 in progress

Associazione Italiana Sommelier (AIS) – Tuscany

Professional sommelier training: viticulture and oenology, tasting technique and sensory analysis, Italian and international wine regions, food and wine pairing, wine service.

PROJECTS

Knowledge Genome Orchestrator

Personal project

A distributed, encrypted, multi-domain knowledge base in which an LLM agent incrementally builds and maintains a persistent wiki, instead of re-deriving knowledge at query time (no vector database, no embedding pipeline, no retrieval server).

- Two-phase agent architecture with an explicit trust boundary: the agent performs semantic work only, a deterministic and tested post-processor validates its output before anything touches the filesystem.
- Per-genome AES-256 encryption (git-crypt) with runtime key injection from a password manager, so the key never touches disk.
- Human review gate through pull requests; deterministic test suite with no dependency on model, GPU or network.
- Multi-domain architecture based on Git submodules, orchestrated via webhooks and local LLM inference.

NETKIT — network field audit toolkit

Personal project

A terminal application for professional network auditing, designed for consultants and freelancers, in which the legal layer is a first-class component rather than an afterthought.

- Digital consent flow producing a non-repudiable SHA-256 hash of agreement, timestamp and scope.
- Fail-closed “scope guard” validating every IP and subnet in real time, with time windows and exclusion lists.
- Alignment with NIS2 and GDPR; local-first AI reporting so audit data never leaves the machine.
- Built on Bun and OpenTUI with strictly typed TypeScript (branded types, result pattern) and SQLite.

Self-hosted infrastructure

Personal project

A fully containerised self-hosted stack built with Docker Compose, deliberately chosen over Kubernetes for operational simplicity, fast recovery and maintainability.

- Reverse proxy and SSL management, private Git hosting (Forgejo), workflow automation (n8n with PostgreSQL and Redis), secure external access via Cloudflare Tunnel.
- Local LLM runtime (Ollama) with NVIDIA GPU acceleration; infrastructure-as-code and reproducible, independently deployable stacks.

SKILLS

Front-end and web

TypeScript · Angular · CSS · JavaScript · implement front-end website design · design user interface · World Wide Web Consortium standards · use markup languages

AI and emerging technologies

principles of artificial intelligence · emergent technologies · use automatic programming

Programming and software engineering

use software design patterns · develop software prototype · object-oriented modelling · use object-oriented programming · use software libraries · debug software

Security and compliance

ICT security legislation · NIS2 and GDPR alignment · digital forensics fundamentals

Infrastructure and tools

tools for software configuration management · Git · Docker · Forgejo · n8n · Ollama · Proxmox

Client and consulting

identify customer requirements · propose ICT solutions to business problems · collect customer feedback on applications · use online tools to collaborate · adapt to changes in technological development plans · keep up with the latest information systems solutions

LANGUAGE SKILLS

Mother tongue

Italian

English

Certified C1 — Program for Intensive English (level 6 of 7), IUPUI, Indianapolis, October 2022.

LISTENING	READING	WRITING	SPOKEN PRODUCTION	SPOKEN INTERACTION
C1	C1	C1	C1	C1

PUBLICATIONS

2022

Progressive Web Applications and Web Operating Systems

Bachelor thesis, University of Florence. Supervisor: Prof. Rosario Pugliese; co-supervisor: Prof. Francesco Tiezzi.

NETWORKS AND MEMBERSHIPS

1 Oct 2025 – Current

Associazione Italiana Sommelier (AIS) — Tuscany

DRIVING LICENCE

11 Jul 2016 – 27 Mar 2034

Category B

9 Sep 2023 – 27 Mar 2034

Category A

HOBBIES AND INTERESTS

Agriculture, food and beverage culture

Sustainable, non-intensive agriculture and quality food supply chains (Slow Food). Director of a family-run agricultural business managing farmland and rural property in Umbria. Viticultural biodiversity through G.R.A.S.P.O., the ampelographic research association recovering ancient and rare grape varieties. Wine culture (AIS member, sommelier in training). Specialty coffee.

Sustainable development, finance and society

Ethical finance, microcredit and ESG investing. Geopolitics, accessibility and digital rights.